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# **Software Requirements Specification**

for

## **<Smart Class and Halls Vacancy Sharing System>**

Version 1.0 approved

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## Revision History

| Name | Date | Reason For Changes | Version |
|------|------|--------------------|---------|
|      |      |                    | 1.0     |
|      |      |                    |         |

# 1 Introduction

## 1.1 Purpose

This document presents the Software Requirements Specification (SRS) for the Smart Class and Halls Vacancy Sharing System. The purpose of this document is to define the functional and non-functional requirements of the system that manages classroom bookings, seminar hall reservations, and campus event scheduling. This SRS describes the complete system including booking management, approval workflows, event proposal submission, and administrative controls. It serves as a reference for developers, stakeholders, and administrators involved in the system development and maintenance.

## 1.2 Document Conventions

This document follows the IEEE standard format for Software Requirements Specification. Section headings are numbered hierarchically. Functional requirements are identified using unique identifiers (e.g., REQ\_01, REQ\_02). All requirements stated in this document are considered mandatory unless otherwise specified. Italicized text is used where necessary to emphasize important concepts.

## 1.3 Intended Audience and Reading Suggestions

This document is intended for developers, project managers, faculty members, system administrators, and stakeholders of the institution. Developers should focus on the functional and interface requirements. Administrators should review system features and security requirements. Faculty members and event organizers should refer to system features related to booking and event management. It is recommended to read the document sequentially, beginning with the Introduction and Overall Description before proceeding to detailed system features.

## 1.4 Product Scope

The Smart Class and Halls Vacancy Sharing System is a campus management solution designed to simplify classroom and seminar hall booking processes. The system allows faculty members and class representatives to request classroom bookings, submit event proposals, and manage academic events. It enables venue administrators to approve or reject booking requests while ensuring no scheduling conflicts occur. The system improves transparency, reduces manual coordination, and ensures efficient utilization of institutional resources.

## 1.5 References

- Use Case Document – Group 15, Version 1.0, February 2026.
- IEEE Software Requirements Specification Template.

## 2 Overall Description

### 2.1 Product Perspective

The Smart Class and Halls Vacancy Sharing System is a web-based application developed to automate the management of classroom and seminar hall bookings within an educational institution. It replaces traditional manual booking processes and reduces scheduling conflicts. The system interacts with users such as Faculty, Class Representatives (CR), Event Organizers, Classroom Admins, Seminar Hall Admins, and System Admins. It maintains centralized data storage for venue details, booking requests, and event records.

### 2.2 Product Functions

The system provides the following major functions:

- Allow faculty and CR to book classrooms.
- Allow submission of event proposals.
- Allow booking of seminar halls for events.
- Enable admins to approve or reject booking requests.
- Provide event modification and cancellation features.
- Maintain real-time venue availability checking.
- Manage user accounts and assign roles.
- Generate system usage reports.

### 2.3 User Classes and Characteristics

The system includes the following user classes:

- **Faculty:** Can book classrooms, request extra classes, and submit event proposals.
- **Class Representative (CR):** Can book classrooms and submit event proposals.
- **Event Organizer:** Can create, modify, or cancel events.
- **Classroom Admin:** Approves or rejects classroom booking requests.
- **Seminar Hall Admin:** Approves or rejects seminar hall booking requests.
- **System Admin:** Manages venues, user accounts, and system records.

### 2.4 Operating Environment

The system operates in a web-based environment and can be accessed through modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. It supports desktop and laptop devices running Windows, macOS, or Linux operating systems. The system requires an active internet connection for real-time booking and notification features.

## 2.5 Design and Implementation Constraints

The system must ensure no double booking of venues. It must follow institutional access control policies. The system should be scalable to support multiple concurrent users during peak academic periods. Data security and privacy must be maintained. The system should be developed using reliable web technologies and a centralized database management system.

## 2.6 User Documentation

User documentation will include:

- User manual describing booking procedures.
- Administrator guide for approval workflows.
- Online help support within the system interface.

## 2.7 Assumptions and Dependencies

- All classrooms and seminar halls are pre-registered in the system.
- Users are assigned appropriate roles by the System Admin.
- The institution provides stable internet connectivity.
- The system depends on a centralized database for storing booking and event information.

# 3 External Interface Requirements

## 3.1 User Interfaces

The Smart Class and Halls Vacancy Sharing System provides a web-based graphical user interface (GUI) accessible through standard web browsers. The interface is designed to be simple, user-friendly, and role-based.

The main user interface components include:

- **Login Page:** Allows users to enter username and password for authentication.
- **Dashboard:** Displays options based on user roles (Faculty, CR, Admin, Event Organizer).
- **Booking Form:** Enables users to select venue, date, time slot, and purpose.
- **Event Proposal Form:** Allows submission of event details including event name, description, expected participants, and preferred venue.
- **Approval Interface:** Allows venue admins to review, approve, or reject booking requests.
- **System Management Interface:** Enables System Admin to manage users and venue details.

All pages shall include navigation menus and logout functionality. Error messages shall be clearly displayed for invalid inputs or scheduling conflicts.

### 3.2 Hardware Interfaces

The system does not require specialized hardware. It can operate on:

- Desktop Computers
- Laptops
- Institutional Computer Lab Systems

Minimum hardware requirements include:

- Processor: Dual Core or higher
- RAM: Minimum 4 GB
- Internet Connectivity

The system does not directly interact with hardware devices such as biometric scanners or physical display boards in the current version.

### 3.3 Software Interfaces

The system interfaces with the following software components:

- Web Browser (Google Chrome, Mozilla Firefox, Microsoft Edge)
- Database Management System (e.g., MySQL or PostgreSQL)
- Web Server (e.g., Apache or Nginx)

The system stores and retrieves data related to users, venues, bookings, and events from a centralized database. It may use backend frameworks and server-side technologies for processing booking requests and approval workflows.

### 3.4 Communications Interfaces

The system requires internet connectivity to function properly. Communication interfaces include:

- HTTP/HTTPS protocol for web communication.
- Email notification system for sending booking confirmations and approval decisions.

All communication shall be secured using HTTPS protocol to ensure data privacy and protection. Notification messages shall be generated automatically when booking status changes or event modifications occur.

## **4 System Features**

### **4.1 Book Classroom**

#### **4.1.1 Description and Priority**

This feature allows Faculty members and Class Representatives (CR) to request booking of a classroom for regular classes, extra classes, or academic events. The system checks real-time availability before submitting the request.

Priority: High

#### **4.1.2 Stimulus/Response Sequences**

1. User selects the “Book Classroom” option.
2. System displays available classrooms.
3. User selects classroom, date, and time slot.
4. System checks classroom availability.
5. System displays availability status.
6. User enters purpose of booking.
7. User submits the booking request.
8. System stores request with status “Pending”.
9. System forwards request to Classroom Admin.
10. System displays confirmation message.

#### **4.1.3 Functional Requirements**

- REQ\_01: The system shall allow Faculty and CR to request classroom booking.
- REQ\_02: The system shall check classroom availability in real time.
- REQ\_03: The system shall prevent double booking of classrooms.
- REQ\_04: The system shall store booking requests with status “Pending”.
- REQ\_05: The system shall notify the respective Classroom Admin.
- REQ\_06: The system shall display appropriate error messages for invalid time slots.



## **4.2 Approve / Reject Classroom Booking**

### **4.2.1 Description and Priority**

This feature allows the Classroom Admin to review pending booking requests and approve or reject them based on availability and scheduling rules.

Priority: High

### **4.2.2 Stimulus/Response Sequences**

1. Admin logs into the system.
2. Admin selects “View Pending Requests”.
3. System displays list of pending bookings.
4. Admin selects a booking request.
5. System displays booking details.
6. Admin reviews date, time, and purpose.
7. Admin selects “Approve” or “Reject”.
8. System updates booking status.
9. System updates classroom schedule if approved.
10. System notifies the requester.

### **4.2.3 Functional Requirements**

- REQ\_07: The system shall allow Classroom Admin to view pending booking requests.
- REQ\_08: The system shall allow Admin to approve booking requests.
- REQ\_09: The system shall allow Admin to reject booking requests with reason.
- REQ\_10: The system shall update classroom schedule upon approval.
- REQ\_11: The system shall notify the user about approval or rejection.
- REQ\_12: The system shall maintain audit logs of approval decisions.

## **4.3 Book Seminar Hall**

### **4.3.1 Description and Priority**

This feature allows Faculty members and Event Organizers to request booking of a seminar hall or auditorium for conducting seminars, workshops, guest lectures, and academic events. The system verifies venue availability before submission.

Priority: High

### **4.3.2 Stimulus/Response Sequences**

1. User selects the “Book Seminar Hall” option.
2. System displays list of available seminar halls.
3. User selects desired hall.
4. User selects date and time slot.
5. System checks hall availability.
6. System displays availability status.
7. User enters event details (Event Name, Purpose, Expected Participants).
8. User submits booking request.
9. System stores request with status “Pending”.
10. System forwards request to Seminar Hall Admin.
11. System displays confirmation message.

### **4.3.3 Functional Requirements**

- REQ\_13: The system shall allow Faculty and Event Organizers to request seminar hall bookings.
- REQ\_14: The system shall check seminar hall availability in real time.
- REQ\_15: The system shall prevent double booking of seminar halls.
- REQ\_16: The system shall allow users to enter detailed event information.
- REQ\_17: The system shall store booking requests with status “Pending”.
- REQ\_18: The system shall notify the respective Seminar Hall Admin.
- REQ\_19: The system shall display validation errors for incomplete event details.

## **4.4 Approve / Reject Seminar Hall Booking**

### **4.4.1 Description and Priority**

This feature allows the Seminar Hall Admin to review pending seminar hall booking requests and approve or reject them based on scheduling constraints and institutional policies.

Priority: High

### **4.4.2 Stimulus/Response Sequences**

1. Admin logs into the system.
2. Admin selects “View Pending Seminar Hall Requests”.
3. System displays list of pending requests.
4. Admin selects a request to review.
5. System displays event details and requested time slot.
6. Admin verifies availability and event details.
7. Admin selects “Approve” or “Reject”.
8. System updates booking status.
9. System updates seminar hall schedule if approved.
10. System sends notification to requester.

### **4.4.3 Functional Requirements**

- REQ\_20: The system shall allow Seminar Hall Admin to view pending booking requests.
- REQ\_21: The system shall allow Admin to approve seminar hall bookings.
- REQ\_22: The system shall allow Admin to reject bookings with a specified reason.
- REQ\_23: The system shall update seminar hall schedule upon approval.
- REQ\_24: The system shall notify the requester of the decision.
- REQ\_25: The system shall maintain audit logs of approval actions.

## **4.5 Submit Event Proposal**

### **4.5.1 Description and Priority**

This feature allows Faculty members, Class Representatives (CR), and Event Organizers to submit event proposals including event details and preferred venue for approval.

Priority: High

### **4.5.2 Stimulus/Response Sequences**

1. User selects “Submit Event Proposal”.
2. System displays event proposal form.
3. User enters event details (Name, Description, Date, Time).
4. User selects preferred venue.
5. System checks preliminary venue availability.
6. User confirms submission.
7. System stores proposal with status “Pending Approval”.
8. System forwards proposal to respective venue admin.
9. System displays confirmation message.

### **4.5.3 Functional Requirements**

- REQ\_26: The system shall allow authorized users to submit event proposals.
- REQ\_27: The system shall allow entry of detailed event information.
- REQ\_28: The system shall verify venue availability before submission.
- REQ\_29: The system shall store proposals with status “Pending Approval”.
- REQ\_30: The system shall notify the respective venue admin.
- REQ\_31: The system shall prevent event scheduling conflicts.

## **4.6 Manage Event (Modify / Cancel Event)**

### **4.6.1 Description and Priority**

This feature allows Event Organizers, Faculty, and CR to modify event details or cancel events before the scheduled date.

Priority: Medium to High

### **4.6.2 Stimulus/Response Sequences**

1. User selects “Manage My Events”.
2. System displays list of user-created events.
3. User selects an event.
4. User modifies event details or selects “Cancel Event”.
5. System checks venue availability if changes are made.
6. User confirms changes.
7. System updates event details or marks event as “Cancelled”.
8. System notifies concerned users and venue admin.

### **4.6.3 Functional Requirements**

- REQ\_32: The system shall allow event creators to modify event details.
- REQ\_33: The system shall allow authorized users to cancel events.
- REQ\_34: The system shall release reserved venue upon cancellation.
- REQ\_35: The system shall check venue availability when event details are modified.
- REQ\_36: The system shall notify affected users of modifications or cancellations.
- REQ\_37: The system shall maintain logs of all event changes.

## **4.7 Manage System (Venue & User Management)**

### **4.7.1 Description and Priority**

This feature allows the System Admin to manage venue records, user accounts, and system configurations including role assignments.

Priority: High

### **4.7.2 Stimulus/Response Sequences**

1. Admin logs into the system.
2. Admin selects “Manage Venues” or “Manage Users”.
3. System displays current records.
4. Admin adds, edits, or removes venue/user information.
5. Admin assigns appropriate user roles.
6. System validates entered information.
7. System updates database records.
8. System confirms successful update.

### **4.7.3 Functional Requirements**

- REQ\_38: The system shall allow System Admin to add or update venue details.
- REQ\_39: The system shall allow System Admin to create, update, or delete user accounts.
- REQ\_40: The system shall allow role assignment to users.
- REQ\_41: The system shall validate input data before saving records.
- REQ\_42: The system shall maintain audit logs for administrative actions.
- REQ\_43: The system shall ensure data consistency during updates.

## **5 Other Nonfunctional Requirements**

### **5.1 Performance Requirements**

- The system shall perform real-time venue availability checking within 3 seconds.
- The system shall support multiple concurrent users during peak academic hours.
- Booking confirmation or rejection notifications shall be generated immediately after admin action.
- The system shall handle at least 200 simultaneous users without performance degradation.
- Page loading time shall not exceed 5 seconds under normal network conditions.

## 5.2 Safety Requirements

- The system shall prevent accidental deletion of venue or booking records.
- The system shall require confirmation before cancelling events.
- The system shall maintain backup records to prevent data loss.
- The system shall ensure that cancelled bookings automatically release reserved time slots.

## 5.3 Security Requirements

- The system shall require user authentication before granting access.
- The system shall implement role-based access control.
- Only authorized admins shall approve or reject bookings.
- The system shall encrypt user passwords in the database.
- The system shall use HTTPS protocol to secure communication.
- The system shall maintain audit logs for all administrative actions.

## 5.4 Software Quality Attributes

- **Usability:** The interface shall be simple and easy to use for faculty and students.
- **Reliability:** The system shall operate without failure during academic sessions.
- **Availability:** The system shall be available 24/7 except during scheduled maintenance.
- **Maintainability:** The system shall allow easy updates and enhancements.
- **Scalability:** The system shall support addition of new venues and increased user load.
- **Portability:** The system shall operate across different operating systems and browsers.

## 5.5 Business Rules

- No classroom or seminar hall shall be double-booked for the same time slot.
- Booking requests must be approved by the respective venue admin before confirmation.
- Only System Admin shall manage venue and user records.
- Only event creators or authorized personnel may modify or cancel events.
- Every user must have an assigned role within the system.

## 6 Other Requirements

### 6.1 Database Requirements

- The system shall maintain a centralized database for storing user accounts, venue details, booking records, and event information.
- Each booking record shall contain user ID, venue ID, date, time slot, purpose, and status.
- Each event record shall store event name, description, organizer details, venue, and schedule.
- The database shall ensure referential integrity and consistency.

### 6.2 Legal Requirements

- The system shall comply with institutional data protection policies.
- User data shall not be shared with unauthorized third parties.

### 6.3 Future Enhancements

- Integration with SMS notification system.
- Resource booking support (Projector, Sound System).
- Automatic recurring booking feature.
- Role-based automated approval for low-priority events.

## Appendix A: Glossary

- **Faculty:** Teaching staff authorized to conduct classes and events.
- **CR (Class Representative):** Student representative authorized to book classrooms.
- **Event Organizer:** User responsible for planning and managing events.
- **Classroom Admin:** Administrator responsible for classroom booking approvals.
- **Seminar Hall Admin:** Administrator responsible for seminar hall approvals.
- **System Admin:** Administrator with full system management privileges.
- **Booking Request:** A request submitted for reserving a venue.
- **Event Proposal:** A formal request to conduct an event within the campus.



## **Appendix B: Analysis Models**

The system may include the following analysis models:

- Use Case Diagram representing user interactions.
- Entity Relationship (ER) Diagram representing database structure.
- Activity Diagrams for booking and approval workflows.
- Sequence Diagrams for event management processes.

## **Appendix C: To Be Determined List**

- Final decision on SMS integration.
- Selection of database management system.
- Maximum supported concurrent user load.
- Future mobile application support.